UCS321 · AI FOR ENGINEERS · MINI PROJECT

Predicting Electric Motor Winding Temperature

A machine-learning virtual sensor that replaces a thermocouple

Problem Statement 6 — General Electric (GE) Power Systems

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1

Why winding temperature is the number that matters

It is the binding constraint on how hard a motor may be driven

A PMSM converts electrical power to mechanical power efficiently, but the loss appears as heat in the stator copper. The winding's enamel insulation has a finite thermal rating. Exceed it and the enamel degrades chemically until adjacent turns short and the machine is destroyed.

180 °C

Class H insulation rating

÷ 2

insulation life per +10 °C

0

production motors with a winding thermocouple

The engineering consequence

A controller that knows this temperature accurately can use the machine's full short-term overload capability.

A controller that does not must apply a fixed conservative derating — and give up torque the motor could physically have delivered.

So the estimate is worth real power density, not just a number on a dashboard.

2

So why not just measure it?

The sensor is practical on a test bench and impractical in a product

ON A TEST BENCH

- Thermocouples bonded directly to the winding
- Cost and fragility are acceptable — it is one machine
- Gives us ground truth to learn from

This dataset comes from exactly such a bench.

IN A PRODUCTION MOTOR

- Sensor must survive winding potential, vibration, thermal cycling
- Leads must exit a sealed housing — a new failure mode
- Cost does not survive a volume bill of materials

So production motors simply do not have one.

OBJECTIVE

Estimate stator winding temperature from the signals a motor drive already measures — accurately enough to use as a thermal protection limit in place of a physical sensor.

Every one of these is already measured for field-oriented control — at zero marginal cost.

i_d , i_q	u_d , u_q	speed
torque	coolant	ambient

Success is defined by deployment, not by a leaderboard.

A model with 1 °C mean error and one 20 °C underestimate still destroys the motor.

So we report max absolute error and % within ±2 °C, not just RMSE.

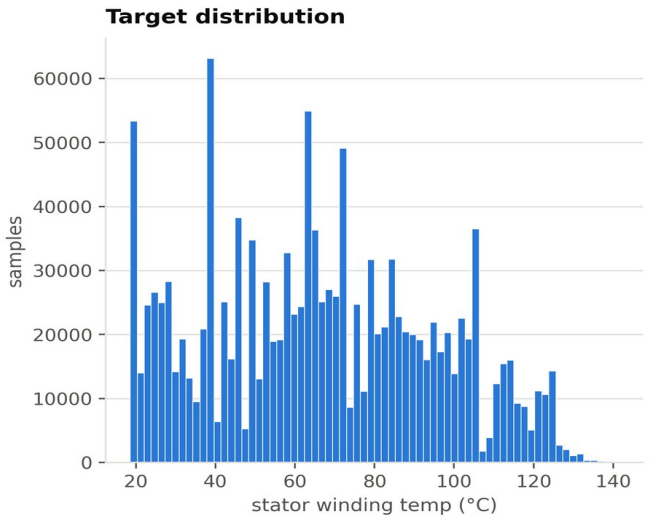
3 The dataset

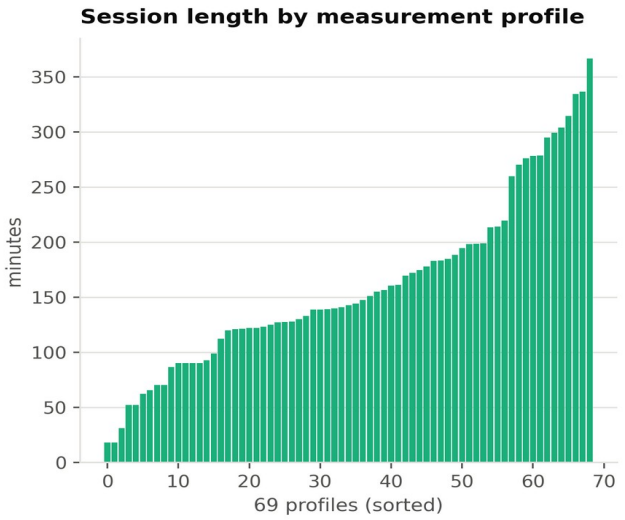
PMSM dynamometer telemetry — Paderborn University (LEA), via Kaggle

1.33 M	2 Hz	69	185 h	0
samples	sampling rate	independent sessions	total recording	missing values

Dataset structure

1,330,816 samples at 2 Hz · 69 independent benchmark sessions · 185 hours total · no missing values





Eight inputs retained

i_d, i_q · u_d, u_q · motor speed · torque · coolant · ambient

Three channels discarded

stator_tooth · stator_yoke · pm

Correlated r > 0.95 with the target — and dropped anyway. Next slide explains why.

4

The most important decision we made

Throwing away the three features that would have scored best

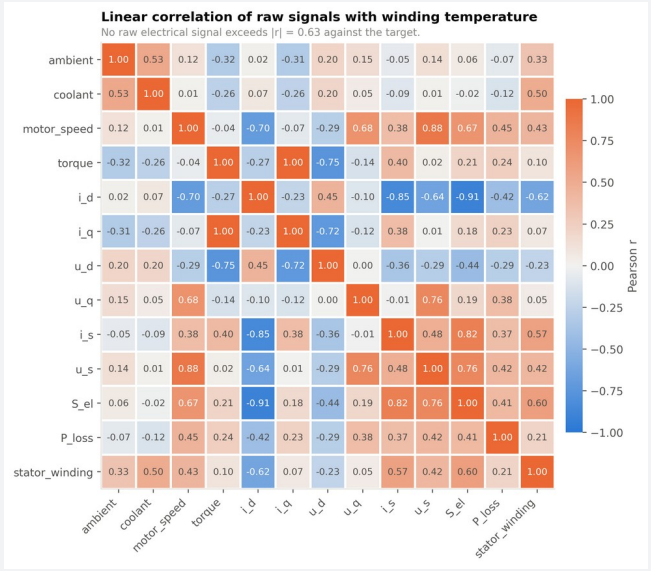
stator_tooth · stator_yoke · pm

correlate with the target at r > 0.95

Including them yields a spectacular R^2. We dropped all three.

- It defeats the objective. Those sensors are exactly the hardware we are trying to remove. A model that needs stator_tooth cannot run on a motor that has neither.
- It is target leakage. Tooth and yoke sit millimetres from the winding in the same iron. The model would be interpolating between adjacent thermocouples — not learning thermodynamics.

What the numbers look like without them



No raw electrical channel exceeds |r| = 0.63.
That ceiling is thermal inertia showing up numerically — and it is what the next slide is about.

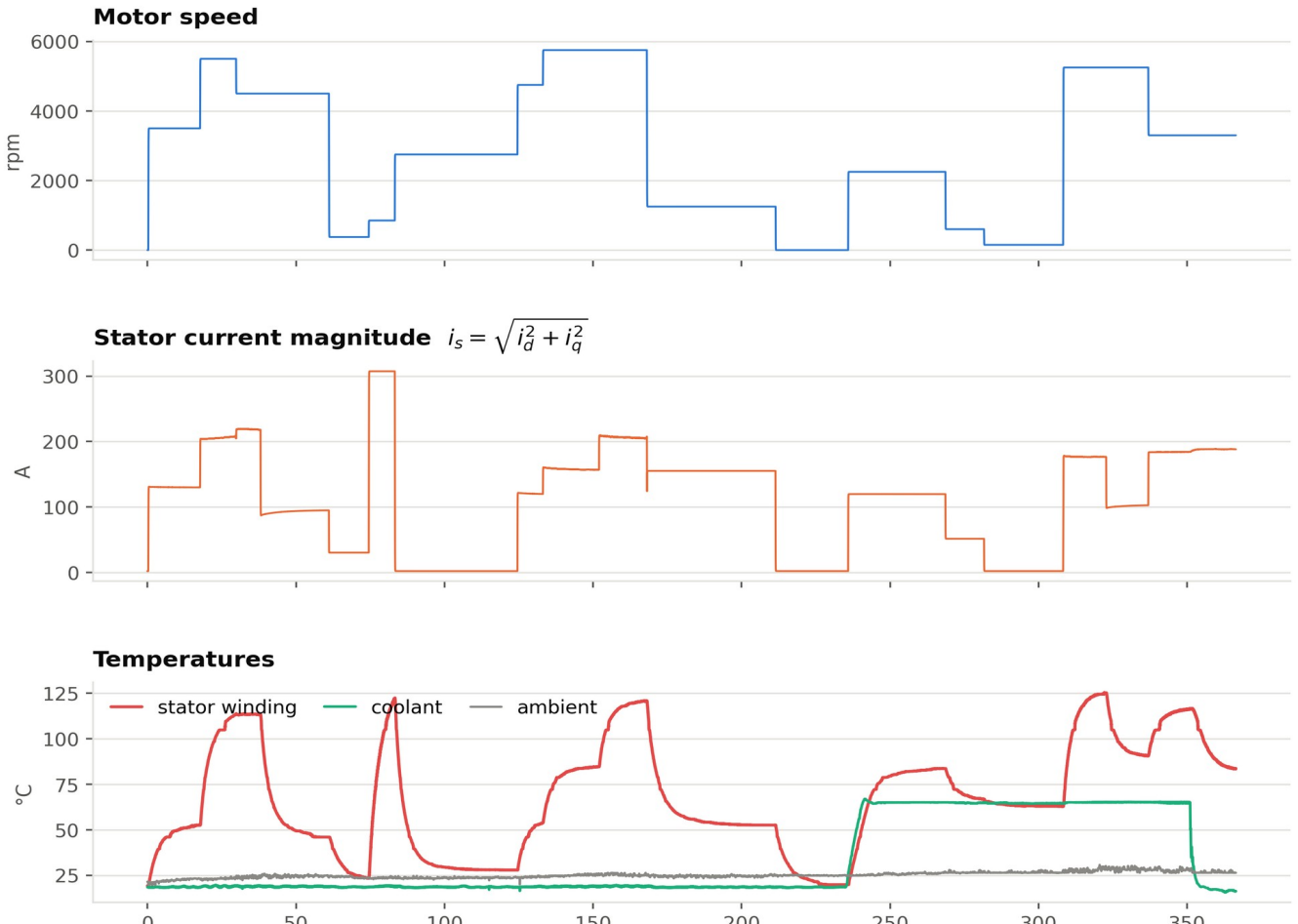
5

The observation the whole project rests on

Current changes in seconds. Temperature answers in minutes.

Thermal response lags the electrical excitation · profile 20

Current swings in seconds; winding temperature answers over minutes. An instantaneous-value model cannot see this.



Why this breaks a naive model

Winding temperature is not a function of the present current. It is a function of the accumulated loss over the recent past.

A model given only instantaneous values is being asked to infer a state variable from its input — which for a system with memory is impossible in principle, not merely hard in practice.

6 From physics to features

The thermal model tells us exactly what transform to apply

LUMPED-PARAMETER THERMAL MODEL

$$C_{th} \cdot d\vartheta/dt = P_{loss}(t) - (\vartheta - \vartheta_{amb}) / R_{th}$$

solving gives

$$\vartheta(t) = (1/\tau) \int \exp(-(t-s)/\tau) \cdot P_{loss}(s)\, ds$$

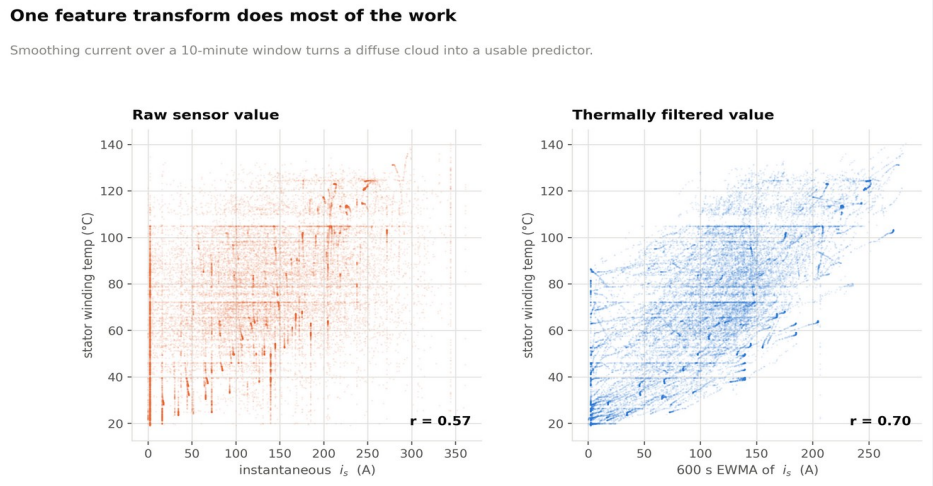
— an exponentially weighted integral of past losses.

which in discrete time is exactly an EWMA: $y[k] = \alpha \cdot x[k] + (1-\alpha) \cdot y[k-1]$

So we compute EWMAs of the loss-related signals at five thermal time constants — 30 s, 120 s, 600 s, 1800 s, 3600 s — because copper, iron and housing respond on different scales.

8 raw channels → **104** engineered features

Does one transform actually help?



Same signal. One 10-minute exponential filter. A diffuse cloud becomes a structured relationship — before any model has been fitted.

7 Why we did not use train_test_split

On 2 Hz data, a random split measures memorisation

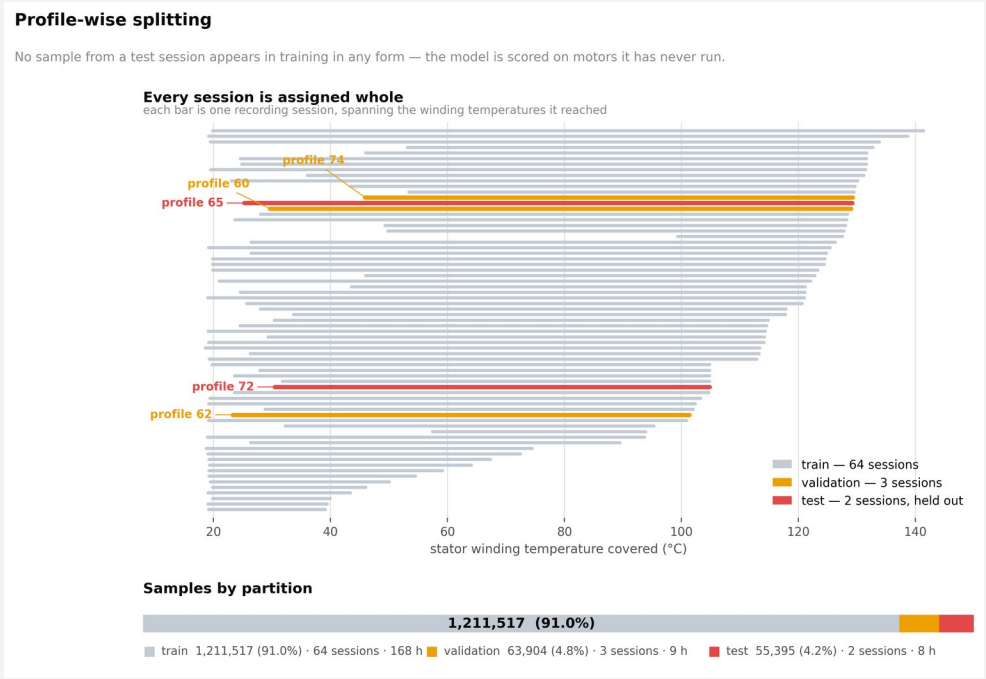
THE TRAP

Samples are 0.5 s apart. A motor's temperature cannot change in half a second, so consecutive rows are near-duplicates. A random split scatters near-identical samples across both sides — the test set is effectively inside the training set.

WHAT WE DID INSTEAD

Hold out entire measurement sessions. The model is evaluated the way it would be used: on a motor run it has never seen.

Train	64 profiles	1,211,517 samples
Validation	3 profiles	63,904 samples
Test	profiles 65 & 72	55,395 samples



The cost of getting this wrong

Random split: R^2 = 0.9995, RMSE 0.66 °C · whole sessions held out: R^2 = 0.9908, RMSE 2.50 °C

Random splitting understates RMSE by ~3.8×.

8

The model ladder

Every increase in complexity has to justify itself against the rung below

1	**Linear regression**	8 raw inputs, OLS	*the honest floor*
2	**Ridge regression**	104 engineered features	*isolates the feature work*
3	**Random Forest**	80 trees, depth 16	*non-parametric reference*
4	**MLP**	2 × 128, early stopping	*smooth approximator*
5	**LightGBM**	743 trees, early-stopped	*strong tabular baseline*

Same data.
Same split.
Same metrics.

All five fitted on the identical training partition and scored on the identical held-out sessions, so the comparison isolates the model and nothing else.

Highlighted rung wins →

THE CENTRAL FINDING

Feature engineering beat every model we tried

Same algorithm. Same data. Same training procedure. Only the inputs changed.

Linear regression on 8 raw inputs

0.5324 — R^2

17.79 °C — RMSE

8.1 % of samples within ±2 °C

→

Linear regression + EWMA features

0.9936 — R^2

2.08 °C — RMSE

81.3 % of samples within ±2 °C

Domain-informed feature engineering outperformed every increase in model sophistication attempted in this project.

9 Full model comparison

Held-out sessions only — profiles 65 and 72, 55,395 samples

Model	RMSE	MAE	R^2	Max err	±2 °C
Linear (raw inputs)	17.79	14.37	0.5324	69.27	8.1 %
Ridge (+ EWMA)	2.08	1.54	0.9936	26.19	71.7 %
Random Forest	3.60	2.49	0.9808	15.56	56.8 %
MLP (2 × 128)	1.67	1.26	0.9959	9.84	81.3 %
LightGBM	2.50	1.85	0.9908	11.05	65.8 %
MLP — tuned	**1.54**	**1.10**	**0.9965**	**11.32**	**84.2 %**

The three metrics do not rank the models identically — see the next slide.

Tuning gained RMSE 1.67 → 1.54 °C, MAE 1.26 → 1.10 °C

Tuning cost worst case 9.84 → 11.32 °C — the search optimised RMSE

Selected model

MLP 2 × 128

tuned

1.54 °C

RMSE

0.9965

R^2

11.32 °C

worst-case error

84.2 %

within ±2 °C

10

Three results worth explaining

The interpretation matters more than the table

1

The neural net beat both tree ensembles.

Tree ensembles are piecewise-constant — a prediction is the mean of the training samples in a leaf, so they cannot produce a value outside the range they saw. Held-out sessions contain operating points the 64 training profiles never visit. An MLP is smooth and continues the trend. Under genuine extrapolation, smoothness is the decisive property.

2

Ridge regression beat LightGBM.

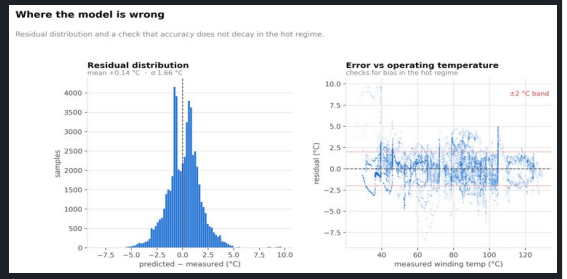
Once the features encode the thermal physics, the residual relationship is close to linear. A low-variance estimator then generalises across sessions better than a high-capacity one, which fits session-specific structure that does not transfer.

3

RMSE and worst-case error disagree.

Ridge attains RMSE 2.08 °C — better than LightGBM's 2.50 °C — yet its worst case is 26.19 °C against 11.05 °C. For a thermal limit the second column governs. A model chosen on RMSE alone would have been the wrong choice.

Residual check



Error does not grow with operating temperature.

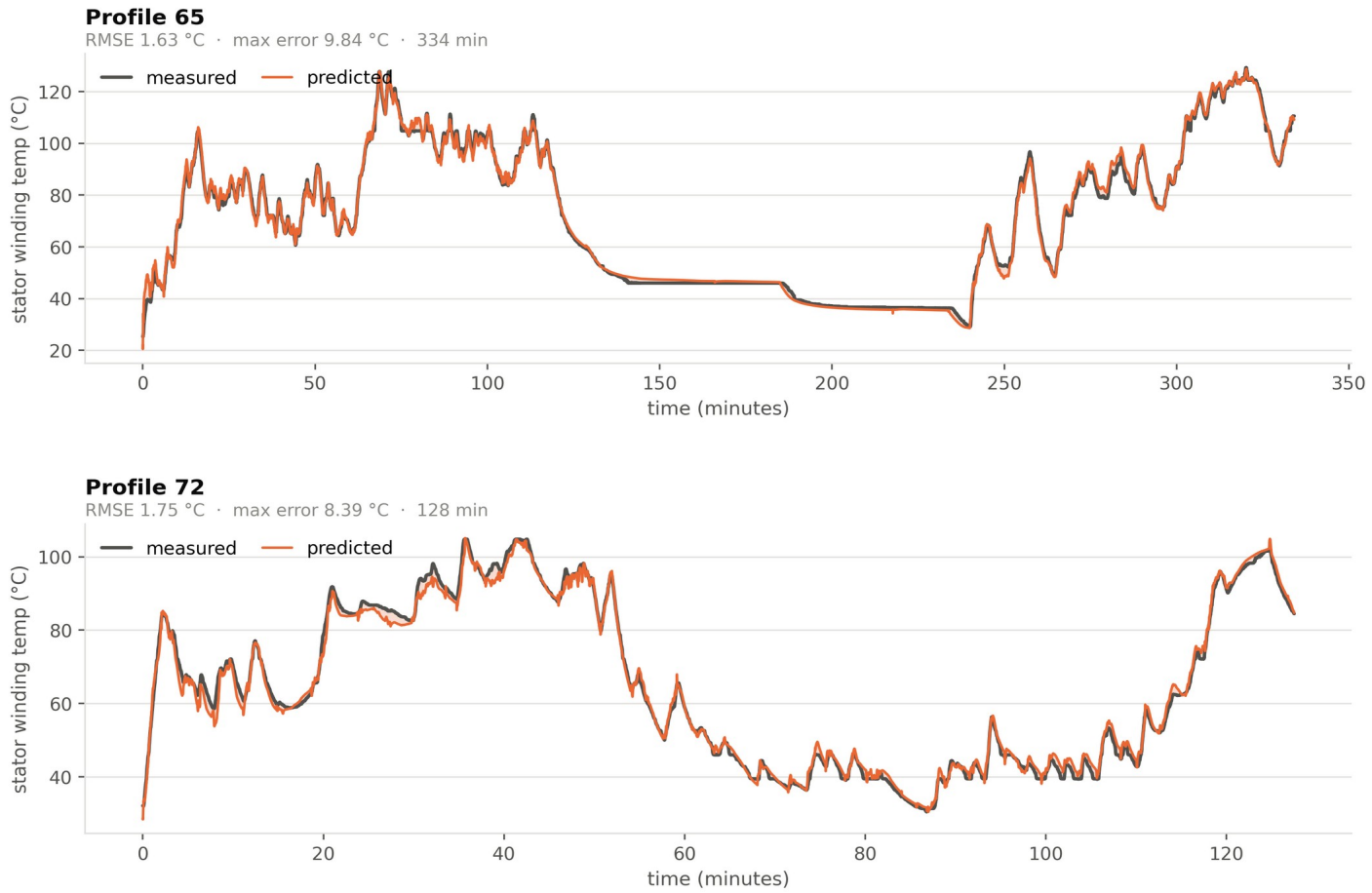
That is the check that matters: a model accurate when cool and poor when hot would be useless, because the hot regime is the only one where the estimate is consulted.

11

Tracking on motors the model has never seen

Neither session appeared in training in any form

MLP (2 x 128): predicted vs measured on unseen motor sessions



What to look at

- Rapid transients under aggressive load cycling are tracked
- Slow exponential cool-down is tracked
- Largest deviations sit at sharp transitions

That lag at step changes is the expected signature of a filter-based feature set meeting a change faster than its shortest time constant.

12

Two studies that test whether the result is real

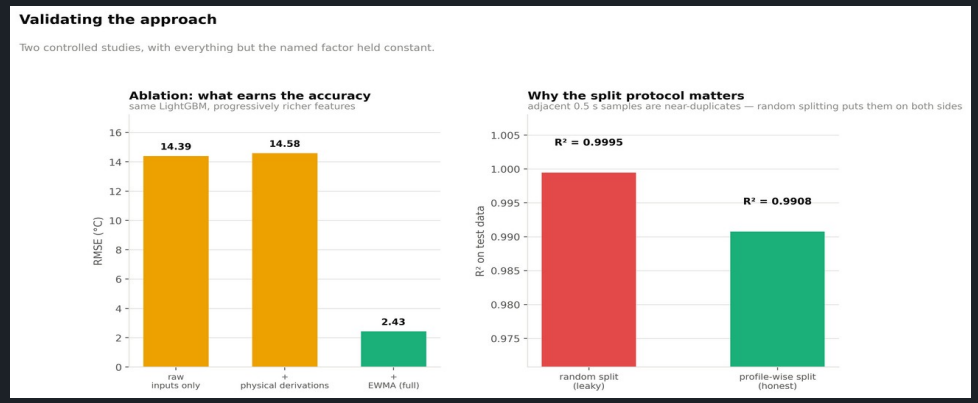
Ablation and leakage — both with everything else held constant

ABLATION — where does the accuracy come from?

	RMSE	R^2
8 raw input channels	14.39	0.6942
+ instantaneous physical terms	14.58	0.6860
+ EWMA thermal history	**2.43**	**0.9913**

Instantaneous physical terms buy nothing on their own — the small regression is within run-to-run variation. Only thermal history moves the result. The gain is not from having more columns.

LEAKAGE — the cost of the wrong split



Same model, same features, only the partitioning changes. Random splitting reports $R^2 = 0.9995$ — a figure describing the model's ability to reproduce samples it has already seen in near-identical form.

For time-series sampled faster than the system's dynamics, grouped splitting is a correctness requirement, not a refinement.

13

Tuning, without undoing the split discipline

RandomizedSearchCV under GroupKFold grouped on profile_id

THE CONSTRAINT THAT MATTERS

A candidate is never scored on a session it was fitted on.

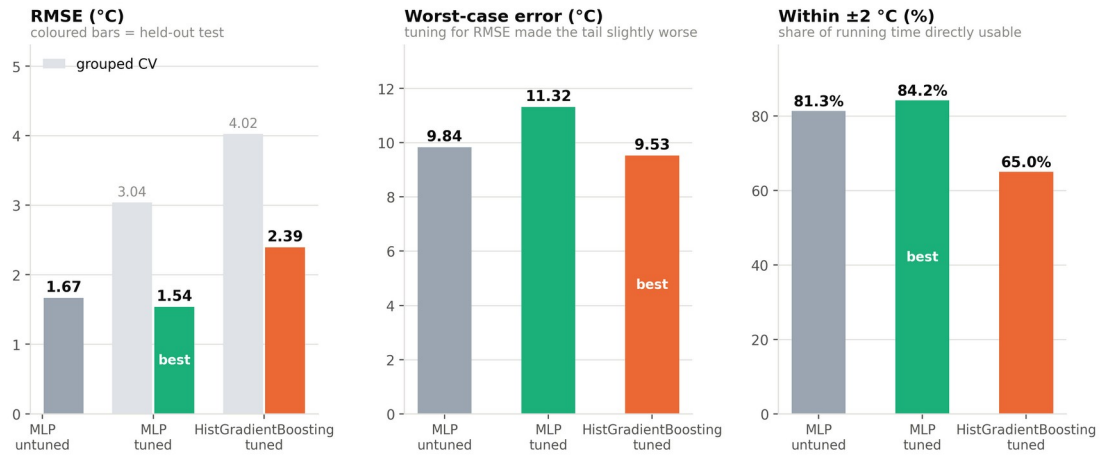
A plain KFold inside the search would reintroduce exactly the leakage the outer split removes — and invisibly, because the CV score would simply look better.

18	3	19
candidates	grouped folds	minutes of search

Selected: hidden (128, 128) · alpha 0.01 · lr 0.001

What tuning bought

RandomizedSearchCV under GroupKFold on profile_id — RMSE and MAE improved; worst-case error did not.



Tuning improved RMSE by ~8 % and MAE by ~13 %. It did not improve worst-case error — that moved from 9.84 °C to 11.32 °C, because the search optimised RMSE and nothing constrained the tail.

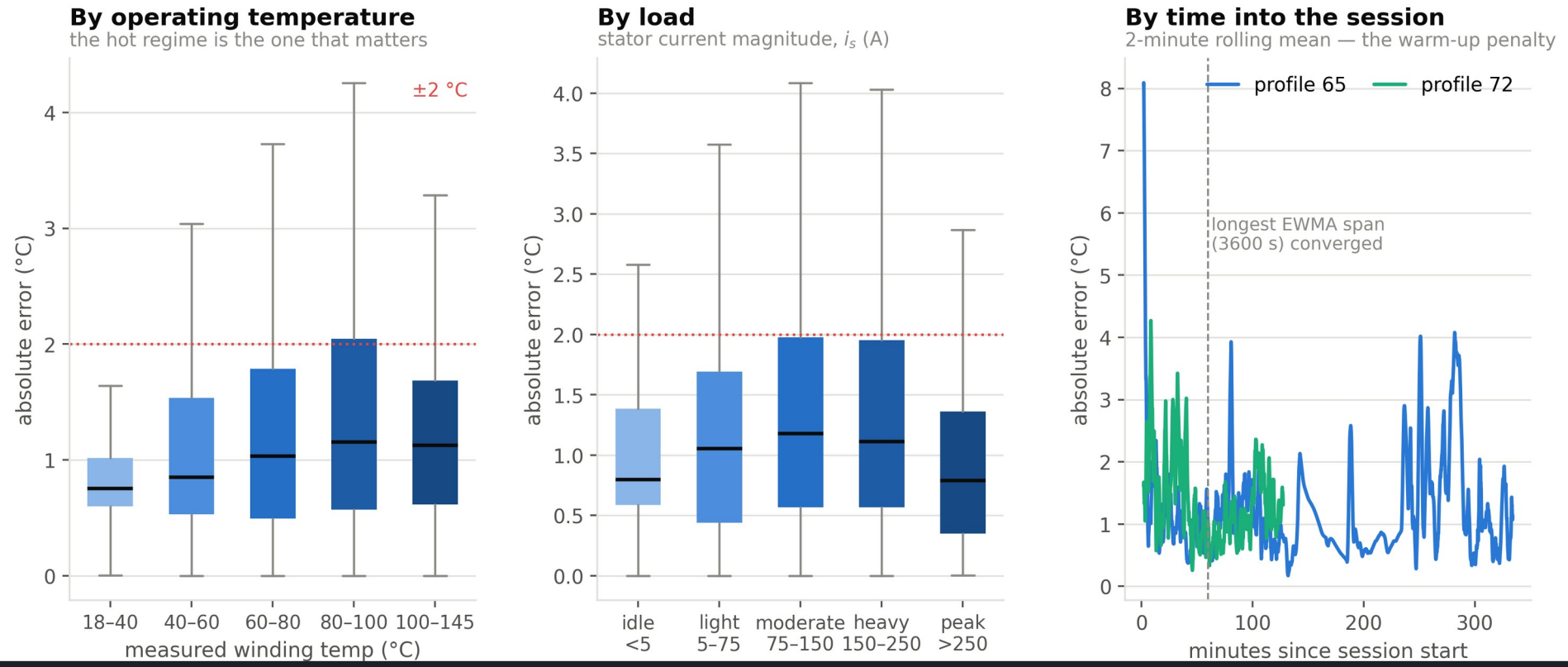
14

Where the error actually lives

Broken down three ways on the held-out sessions

Where the error actually lives

Median error stays near 1 °C in every regime. The tail widens under load, and the worst errors sit in the first minutes, while the long filters are still converging.

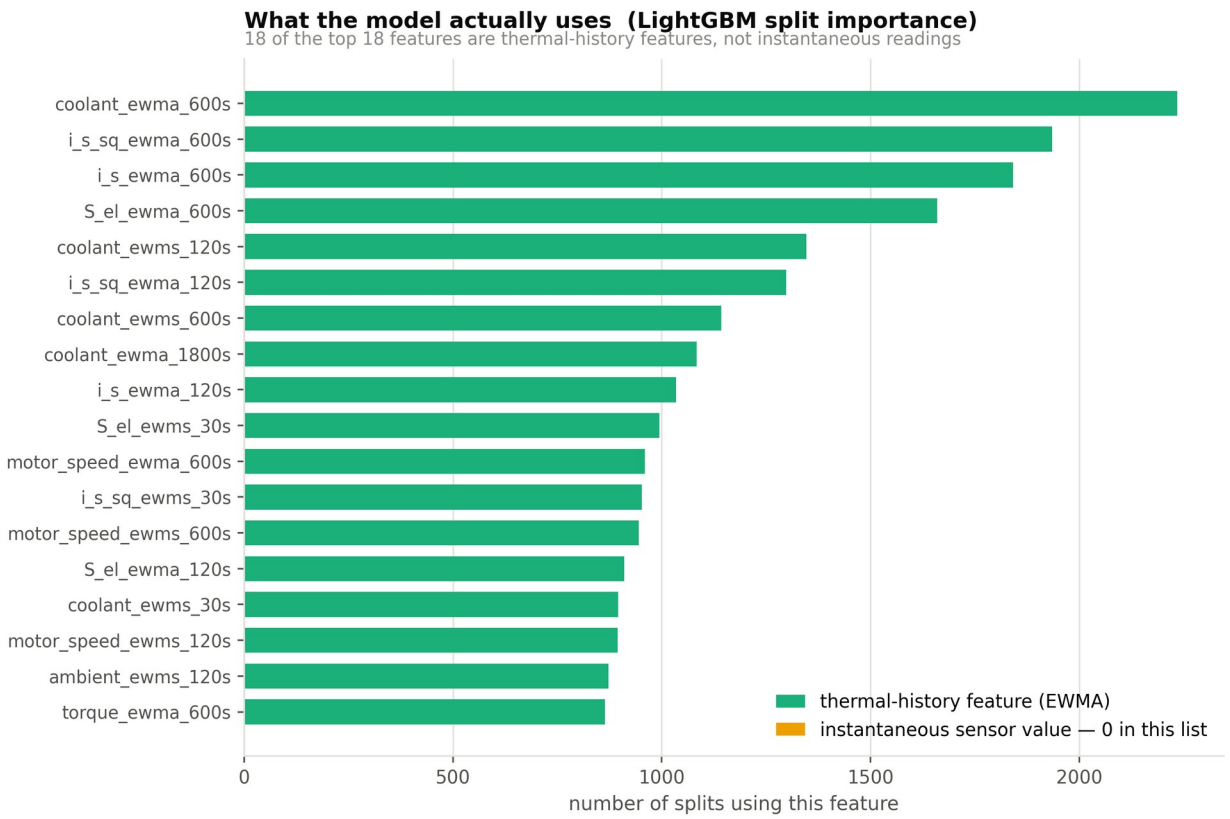


Median error holds near 1 °C across every temperature and load band — accuracy does not decay when the motor is hot or working hard. The real weakness is the third panel: the worst errors sit in the first minutes, while the 1800 s and 3600 s filters are still converging.

15

The model learned the physics we designed for

Strongest evidence that the result is not an artefact



All top-15 features are thermal history

coolant_ewma_600s

the heat-sink boundary condition

i_s^2_ewma_600s

the copper-loss term, $P = I^2R$

These are exactly the two terms the lumped-parameter thermal model on slide 8 identifies as dominant.

Given no information about motor thermodynamics beyond which signals to filter, the model independently concentrated on them.

16

It is cheap enough to run on the drive

Which is what makes it a replacement for a thermocouple, not an offline analysis

~60 k

MAC/s at the 2 Hz update rate

90

state registers for the entire feature extractor

1

multiply-add per EWMA feature

16-bit

sufficient output precision (Q16.16 accumulators)

WHY THE FEATURE EXTRACTOR IS HARDWARE-FRIENDLY

Each EWMA is a single-pole IIR filter: one multiply, one add, one register. No windowing buffer, no stored history.

A conventional one-hour sliding-window statistic at 2 Hz would need 7,200 buffered samples per signal. The recursive form needs one.

HONEST LIMITATIONS

- Worst case is still 9.84 °C — a deployed limit needs a margin, just a smaller one than a fixed derating
- Trained on one machine on one bench; a different design needs retraining or transfer learning
- Error is largest at sharp load transitions

CONCLUSIONS

What we take away

1. **The objective was met.**

   RMSE 1.54 °C, R^2 0.9965, 84.2 % within ±2 °C on motor sessions never seen during training.

2. **Feature engineering, not model selection, produced the result.**

   An unchanged linear regression improves from R^2 0.53 to 0.99 once its inputs encode thermal history.

3. **The evaluation protocol changes the answer by 4×.**

   Profile-wise splitting is a correctness requirement on data of this kind, not a refinement.

4. **A smooth model generalises better than tree ensembles here.**

   Held-out sessions demand extrapolation, and piecewise-constant models cannot extrapolate.

5. **The model learned the intended physics.**

   Its dominant features are the coolant boundary condition and the copper-loss term.

Future work: sequence models that learn the time constants · multi-output prediction of all four temperatures · physics-informed loss · fixed-point RTL implementation of the EWMA extractor

Anticipated questions

Backup slide — not presented unless asked

Why not just use a lumped-parameter thermal model?

They exist and work, but need per-machine parameter identification on a bench. We reach comparable accuracy directly from logged data, and our features are that model's analytical solution anyway.

Wouldn't keeping stator_tooth give better accuracy?

Yes — and the result would be worthless. Those sensors are the hardware we are replacing; a production motor has none of them.

Why does the MLP beat gradient boosting?

Trees predict the mean of a leaf, so they cannot output values beyond the training range. Held-out sessions require extrapolation; a smooth model handles it.

Is $R^2 = 0.996$ not suspiciously high?

It would be under a random split — we show that gives 0.9995 and is meaningless. Ours is on entirely held-out sessions, and we report the 9.84 °C worst case alongside it.

How did you choose the five EWMA spans?

From the physical time constants: copper responds in tens of seconds, iron in minutes, housing and coolant in tens of minutes. We bracketed that range and let the model weight them.

What would you do with more time?

A temporal CNN or LSTM that learns the time constants instead of having them fixed, and a physics-informed loss enforcing the thermal ODE.